A Very Brief Introduction to Java

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Chapter Outline

Class, Object and Constructor

Exception Handling

Class, Object and Constructor

Class and Object

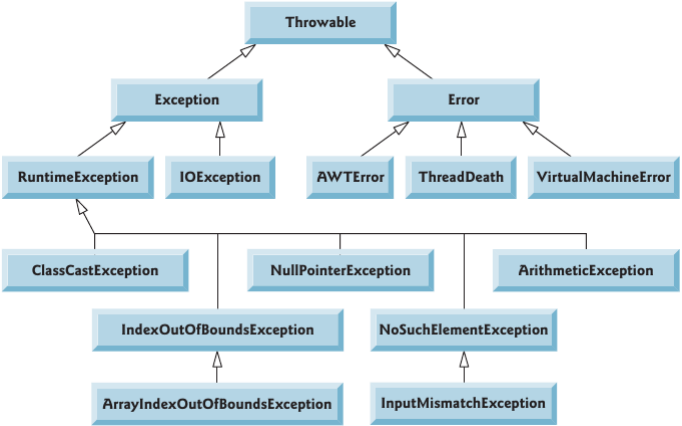
A class is a blueprint or template for creating objects, while an object is a specific, real-world instance of a class.

```
public class HelloWorld {
  public static void main(String[] args) {
    System.out.println("Hello, World");
  }
}
```

Class, Object and Constructor

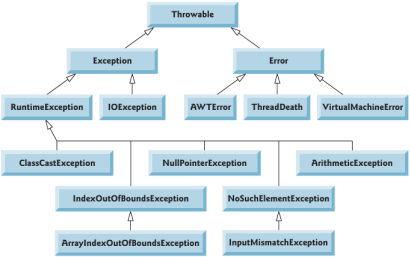
```java
public class CarDemo {
  public static void main(String[] args) {
    // 1. Create the first object of the Car class
    Car car1 = new Car();
     car1.brand = "Toyota";
    car1.year = 2022;

    System.out.println("Details of car1:");
    car1.displayInfo();

    }
}
// Define a separate class (blueprint)
class Car {
  String brand;
  int year;

  void displayInfo() { // method
    System.out.println("Brand: " + brand);
    System.out.println("Year: " + year);
  }
}
```

Exceptions in Java



Exceptions in Java

- Java distinguishes between checked exceptions and unchecked exceptions.
- Java compiler **enforces** special requirements for **checked exception**
- All **I/O operations** are checked exception.
- Unchecked exceptions are not enforced
- Exceptions are handled using `try{ }catch{}` block



Try Catch block

```java
public class TryCatchExample {
  public static void main(String[] args) {
    try {
      int dividend = 10;
      int divisor = 0;
      int result = dividend / divisor;
      System.out.println("Result: " + result); // This line will not execute if an exception occurs
    } catch (ArithmeticException e) {
     System.out.println("Error: Cannot divide a number by zero.");
      // You can also print the technical exception details
      // System.out.println("Exception details: " + e.getMessage());
    }
    System.out.println("Program continues after the try-catch block.");
  }
}
```